	Course Name: Advanced Web Technology	EXPERIMENT NO. 10
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
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Objective: Implement a web-app using Django framework

1. Design a 'Hello World' Web-application using Django framework *views.py*

```
from django.shortcuts import render
```

```
# Create your views here.
```

```
from django.http import HttpResponse
```

```
def hello_world(request):
    return HttpResponse("Hello World!")
```

urls.py

```
"""
```

URL configuration for Lab_10 project.

The `urlpatterns` list routes URLs to views. For more information please see:

<https://docs.djangoproject.com/en/4.2/topics/http/urls/>

Examples:

Function views

1. Add an import: from my_app import views
2. Add a URL to urlpatterns: path("", views.home, name='home')

Class-based views

1. Add an import: from other_app.views import Home
2. Add a URL to urlpatterns: path("", Home.as_view(), name='home')

Including another URLconf

1. Import the include() function: from django.urls import include, path
2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))

```
"""
```

```
from django.urls import path
```

```
from hello_world_app import views
```

```
urlpatterns = [
    path('hello/', views.hello_world, name='hello_world'),
]
```

```
PS D:\SEM-6\Advanced Web Technology> pip install django
Requirement already satisfied: django in c:\python39\lib\site-packages (4.2.11)
Requirement already satisfied: asgiref<4,>=3.6.0 in c:\python39\lib\site-packages (from django) (3.8.1)
Requirement already satisfied: sqlparse>=0.3.1 in c:\python39\lib\site-packages (from django) (0.5.0)
Requirement already satisfied: tzdata in c:\python39\lib\site-packages (from django) (2024.1)
Requirement already satisfied: typing-extensions>=4 in c:\python39\lib\site-packages (from asgiref<4,>=3.6.0->django) (4.7.1)

[notice] A new release of pip is available: 23.3.1 -> 24.0
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
PS D:\SEM-6\Advanced Web Technology> django-admin startproject Lab_10
PS D:\SEM-6\Advanced Web Technology> cd Lab_10
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py startapp hello_world_app
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:14:38
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[25/Apr/2024 13:14:45] "GET / HTTP/1.1" 200 10664
Not Found: /favicon.ico
[25/Apr/2024 13:14:50] "GET /favicon.ico HTTP/1.1" 404 2110
[25/Apr/2024 13:15:16] "GET / HTTP/1.1" 200 10664
Not Found: /hello/
[25/Apr/2024 13:15:29] "GET /hello/ HTTP/1.1" 404 2095
Not Found: /hello
[25/Apr/2024 13:15:35] "GET /hello HTTP/1.1" 404 2092
Not Found: /hello/
[25/Apr/2024 13:15:47] "GET /hello/ HTTP/1.1" 404 2095
Not Found: /hello/
[25/Apr/2024 13:16:05] "GET /hello/ HTTP/1.1" 404 2095
PS D:\SEM-6\Advanced Web Technology\Lab_10> pip install django
Requirement already satisfied: django in c:\python39\lib\site-packages (4.2.11)
Requirement already satisfied: asgiref<4,>=3.6.0 in c:\python39\lib\site-packages (from django) (3.8.1)
Requirement already satisfied: sqlparse>=0.3.1 in c:\python39\lib\site-packages (from django) (0.5.0)
Requirement already satisfied: tzdata in c:\python39\lib\site-packages (from django) (2024.1)
Requirement already satisfied: typing-extensions>=4 in c:\python39\lib\site-packages (from asgiref<4,>=3.6.0->django) (4.7.1)
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:17:37
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[25/Apr/2024 13:17:43] "GET / HTTP/1.1" 200 10664
Not Found: /hello/
[25/Apr/2024 13:17:54] "GET /hello/ HTTP/1.1" 404 2095
D:\SEM-6\Advanced Web Technology\Lab_10\Lab_10\urls.py changed, reloading.
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:20:55
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

Not Found: /
[25/Apr/2024 13:20:58] "GET / HTTP/1.1" 404 2079
[25/Apr/2024 13:21:08] "GET /hello/ HTTP/1.1" 200 12
```



127.0.0.1:8000/hello/



127.0.0.1:8000/hello/

Hello World!

2. Perform CRUD commands in a designed model using the default SQLite database extension in Django.

models.py

```
from django.db import models
```

```
class MyModel(models.Model):
```

```
    name = models.CharField(max_length=100)
```

```
    description = models.TextField()
```

```
    class Meta:
```

```
        app_label = 'hello_world_app'
```

settings.py

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'hello_world_app'  
]
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py makemigrations
```

```
No changes detected
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py migrate
```

```
Operations to perform:
```

```
  Apply all migrations: admin, auth, contenttypes, sessions
```

```
Running migrations:
```

```
  No migrations to apply.
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py makemigrations hello_world_app
```

```
Migrations for 'hello_world_app':
```

```
  hello_world_app\migrations\0002_remove_mymodel_description.py
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py migrate
```

```
Operations to perform:
```

```
  Apply all migrations: admin, auth, contenttypes, hello_world_app, sessions
```

```
Running migrations:
```

```
  Applying hello_world_app.0002_remove_mymodel_description... OK
```

Output:

View SQL

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> py manage.py sqlmigrate hello_world_a
pp 0001
BEGIN;
--
-- Create model MyModel
--
CREATE TABLE "hello_world_app_mymodel" ("id" integer NOT NULL PRIMARY KEY AUTOINC
REMENT, "name" varchar(100) NOT NULL, "description" text NOT NULL);
COMMIT;
```

Add Records

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py shell
Python 3.9.6 (tags/v3.9.6:db3ff76, Jun 28 2021, 15:26:21) [MSC v.1929 64 bit (AMD
64)]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.17.2 -- An enhanced Interactive Python. Type '?' for help.
```

```
In [1]:
```

```
In [1]: from hello_world_app.models import MyModel

In [2]: MyModel.objects.all()
Out[2]: <QuerySet []>

In [3]: text = hello_world_app(name="Adit Shah")
-----
NameError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 text = hello_world_app(name="Adit Shah")

NameError: name 'hello_world_app' is not defined

In [4]: text = MyModel(name="Adit Shah")

In [5]: text.save()

In [6]: MyModel.objects.all().values()
Out[6]: <QuerySet [{'id': 1, 'name': 'Adit Shah'}]>
```

Update Records

```
In [9]: x = MyModel.objects.all()[0]

In [10]: x.name
Out[10]: 'Adit Shah'

In [11]: x.name = "21BCP419"

In [12]: x.save()

In [13]: MyModel.objects.all().values()
Out[13]: <QuerySet [{'id': 1, 'name': '21BCP419'}]>
```

Delete Records

```
In [15]: x = MyModel.objects.all()[0]

In [16]: x.name
Out[16]: '21BCP419'

In [17]: '21BCP419'
Out[17]: '21BCP419'

In [18]: x.delete()
Out[18]: (1, {'hello_world_app.MyModel': 1})

In [19]: MyModel.objects.all().values()
Out[19]: <QuerySet []>
```

3. Connect your Django web-application to PostgreSQL database and perform CRUD commands.

settings.py

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.postgresql',  
        'NAME': 'user',  
        'USER': 'postgres',  
        'PASSWORD': 'password',  
        'HOST': 'localhost',  
        'PORT': '5432',  
    }  
}
```

Views.py

```
from django.shortcuts import render, redirect
```

```
from .models import Item
```

```
def index(request):
```

```
    items = Item.objects.all()
```

```
    return render(request, 'index.html', {'items': items})
```

```
def create(request):
```

```
    if request.method == 'POST':
```

```
        name = request.POST.get('name')
```

```
        description = request.POST.get('description')
```

```
        Item.objects.create(name=name, description=description)
```

```
        return redirect('index')
```

```
    return render(request, 'create.html')
```

```
def update(request, item_id):
```

```
    item = Item.objects.get(id=item_id)
```

```
    if request.method == 'POST':
```

```
        item.name = request.POST.get('name')
```

```
        item.description = request.POST.get('description')
```

```
        item.save()
```

```
        return redirect('index')

    return render(request, 'update.html', {'item': item})
```

```
def delete(request, item_id):
    item = Item.objects.get(id=item_id)
    item.delete()
    return redirect('index')
```

urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path("", views.index, name='index'),
    path('create/', views.create, name='create'),
    path('update/<int:item_id>/', views.update, name='update'),
    path('delete/<int:item_id>/', views.delete, name='delete'),
]
```

Create.html

```
<h1>Create Item</h1>
<form method="post">
    {% csrf_token %}
    <label for="name">Name:</label>
    <input type="text" name="name" required>
    <br>
    <label for="description">Description:</label>
    <textarea name="description" required></textarea>
    <br>
    <input type="submit" value="Create">
</form>
```

Index.html

```
<h1>Items</h1>

<ul>

    {% for item in items %}

        <li>{{ item.name }} - {{ item.description }}</li>

    {% endfor %}

</ul>

<a href="{% url 'create' %}">Create Item</a>
```

Update.html

```
<h1>Update Item</h1>

<form method="post">

    {% csrf_token %}

    <label for="name">Name:</label>

    <input type="text" name="name" value="{{ item.name }}" required>

    <br>

    <label for="description">Description:</label>

    <textarea name="description" required>{{ item.description }}</textarea>

    <br>

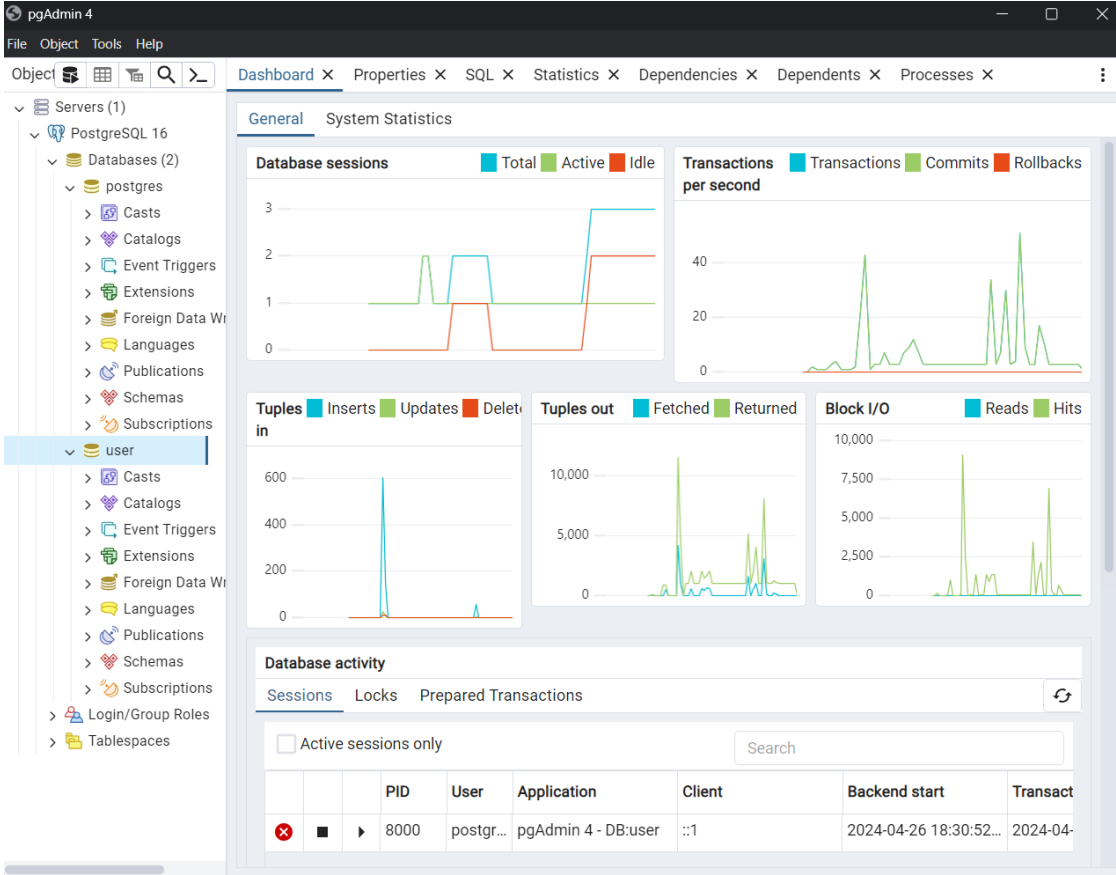
    <input type="submit" value="Update">

</form>
```

Output

```
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py makemigrations
Migrations for 'crud_app':
  crud_app\migrations\0001_initial.py
    - Create model Item
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, crud_app, sessions
Running migrations:
  Applying crud_app.0001_initial... OK
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
April 26, 2024 - 19:02:22
Django version 4.2.11, using settings 'crud_project.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```



Output:

127.0.0.1:8000

127.0.0.1:8000

Items

[Create Item](#)

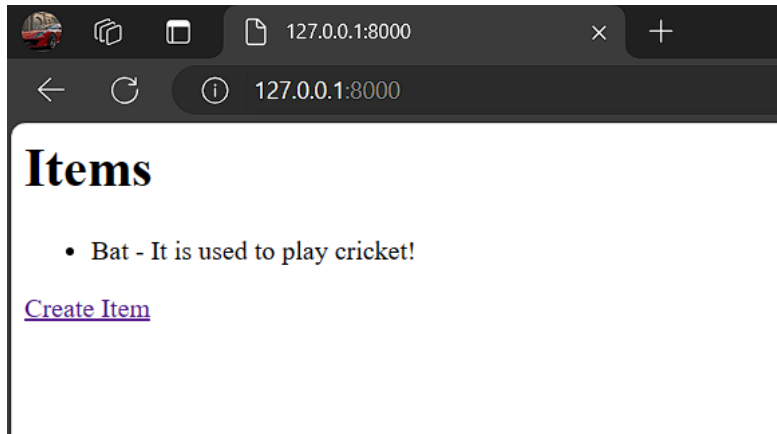
Add Records

Create Item

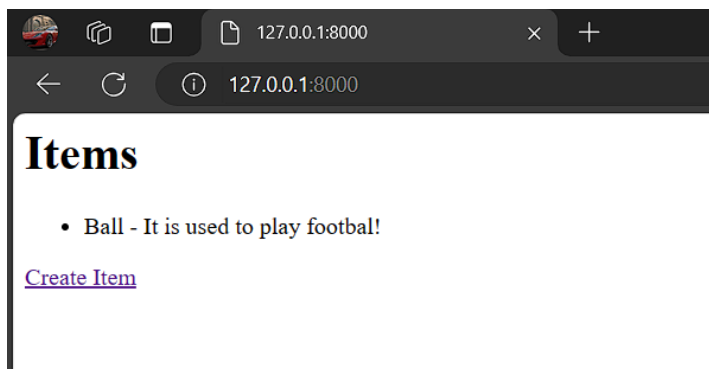
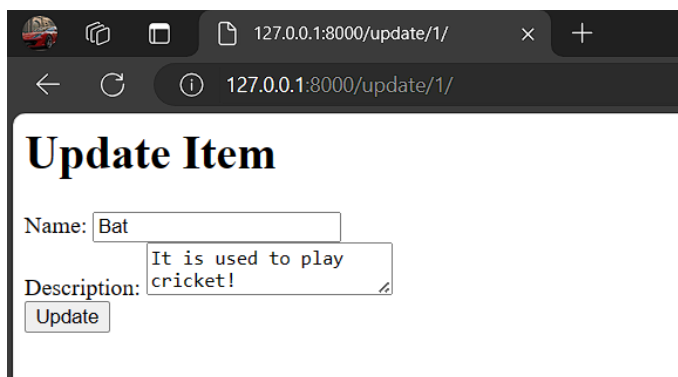
Name:

Description:





Update Records



Delete Records

